MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Electrical Engineering and Computer Science

Problem Set No. 2	6.630 Electromagnetics	Issued: 091108R
Fall Term 2008		Due: 091808R

Reading assignment: Section 1.2-1.3 ; J. A. Kong, "*Electromagnetic Wave Theory*," EMW Publishing.

Problem P2.1

In the United States, broadcasting to the general public is regulated by the Federal Communications Commission (FCC), which allocates frequencies and establishes technical standards. Three general classes of broadcast stations have been established. *Standard broadcast stations* (amplitude modulation-AM) are licensed for operation on channels spaced by 10 kHz and occupying the band from 535 to 1605 kHz. *Frequency modulation* (FM) *broadcast stations* are authorized for operation on 100 allocated channels, each 200 kHz wide, extending consecutively from 201 on 88.1 MHz to channel 300 on 107.9 MHz. *Television broadcast stations* (operating with vestigial-sideband amplitude modulation of the visual carrier and frequency modulation of the aural carrier) are authorized for commercial and educational operation on designated channels 2–83, each 6 MHz wide, extending from 54–806 MHz. (See reference: *Reference Data for Radio Engineers*, Howard W. Sams & Co., ITT, 1983.)

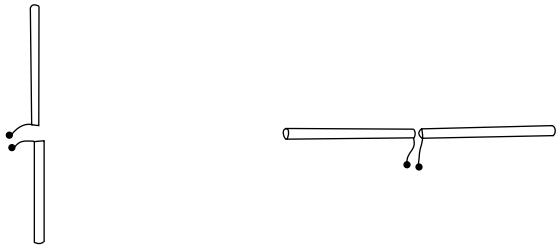
天线

The electromagnetic waves radiated by AM stations have the $\overline{E}$ field perpendicular to the ground and parallel to the antenna towers (vertical polarization). Most FM stations broadcast with circular polarization. For television broadcasting, the $\overline{E}$ field is parallel to the ground (horizontal polarization).

垂直的

The induced current on a receiving antenna is largest when the antenna is aligned with the electric field. Given are two wire antenna configurations. For each type of broadcast, AM, FM, and TV, specify which configuration(s) gives maximum reception

(i) (ii)



Problem P2.2

相量表示

时谐

Obtain the phasor notation of the following time-harmonic functions (if possible):

(a) $V(t) = 6\cos(\omega t + \pi/4)$
(b) $I(t) = -8\sin(\omega t)$
(c) $A(t) = 3\sin(\omega t) - 2\cos(\omega t)$
(d) $C(t) = 6\cos(120\pi t - \pi/2)$
(e) $D(t) = 1 - \cos(\omega t)$
(f) $U(t) = \sin(\omega t + \pi/3)\sin(\omega t + \pi/6)$

Problem P2.3

Consider an electromagnetic wave propagating in the $\hat{z}$-direction with

$$\overline{E} = \hat{x}e_x \cos(kz - \omega t + \psi_x) + \hat{y}e_y \cos(kz - \omega t + \psi_y)$$

where e_x, e_y, ψ_x, and ψ_y are all real numbers.

(a) Let $e_x = 2$, $e_y = 1$, $\psi_x = \pi/2$, $\psi_y = \pi/4$. What is the polarization?

(b) Let $e_x = 1$, $e_y = \psi_x = 0$. This is a linearly polarized wave. Prove that it can be expressed as the superposition of a right-hand circularly polarized wave and a left-hand circularly polarized wave.

(c) Let $e_x = 1$, $\psi_x = \pi/4$, $\psi_y = -\pi/4$, $e_y = 1$. This is a circularly polarized wave. Prove that it can be decomposed into two linearly polarized waves.

Problem P2.4 空间视点

Wave polarization can be viewed by either taking a series of still pictures at several fixed times, called the spatial view point or by making observations at a fixed point in space, called the temporal view point. We define polarization from the temporal view point. Let us now look at polarization from the spatial view point.

Consider an electromagnetic wave with $k = 100\mathrm{K_o}$ propagating in the $\hat{z}$ direction.

时间视点 $$\overline{E}(\overline{r}, t) = E_0[\hat{x}\cos(kz - \omega t) + \hat{y}\sin(kz - \omega t)]$$

What is the wavelength and the polarization of this wave?

From the spatial point of view, by taking a picture at $t = 0$, the tips of the electric field vectors form a helix. Is the helix right-handed or left-handed? What is the pitch(wavelength) of this helix? 螺旋

螺距

Problem P2.5

The Earth receives over all frequency bands about $1.5\,\mathrm{kW/m^2}$ of power from the Sun.

(a) The Earth-Sun distance is 150×10^9 m. How long does it take the sunlight to reach the Earth? 半径

(b) The Earth radius is 6400 km. What is the total power received by the Earth?

(c) Assume the Sun's mass is $2 \times 10^{30}\,\mathrm{kg}$ which converts to radiated energy according to mc^2 at 1 percent efficiency. How long can the Sun radiate at the present level?

(c) The Sun radiates $10^{-20}\ \mathrm{W\,m^{-2}\,Hz^{-1}}$ at 3 GHz. Assuming constant power level over 1 GHz bandwidth, what is the Poynting power density and the corresponding electric field amplitude?

P1.2.3

An electromagnetic wave has spatial frequency $k_o = 100\,\mathrm{K}_o$. Determine the wavelength in meters and the temporal frequency in GHz.

Determine the spatial frequency in unit of K_o for a laser light at wavelength $\lambda = 0.6328\,\mu\mathrm{m}$.

Determine the spatial frequency in unit of K_o for a microwave oven at frequency 2.4 GHz.